



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

Clustering

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github.com/laplacebeltrami

Learning goal

Understand how clustering works

Understand relationship between clustering and statistical inference

Understand state space modeling

Magnetic resonance imaging (MRI)

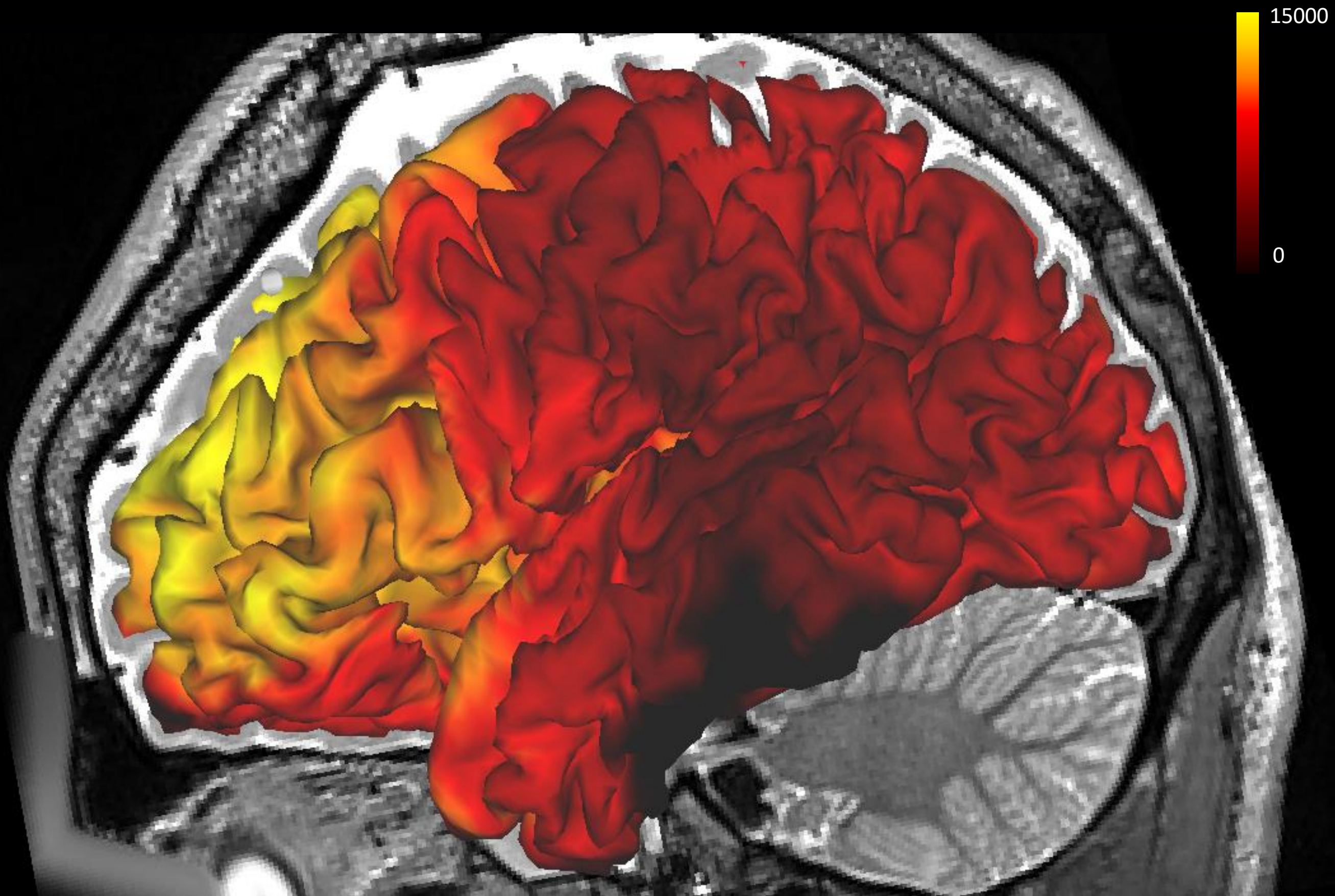


3T GE Discovery X750
Waisman Brain Imaging Laboratory
University of Wisconsin-Madison



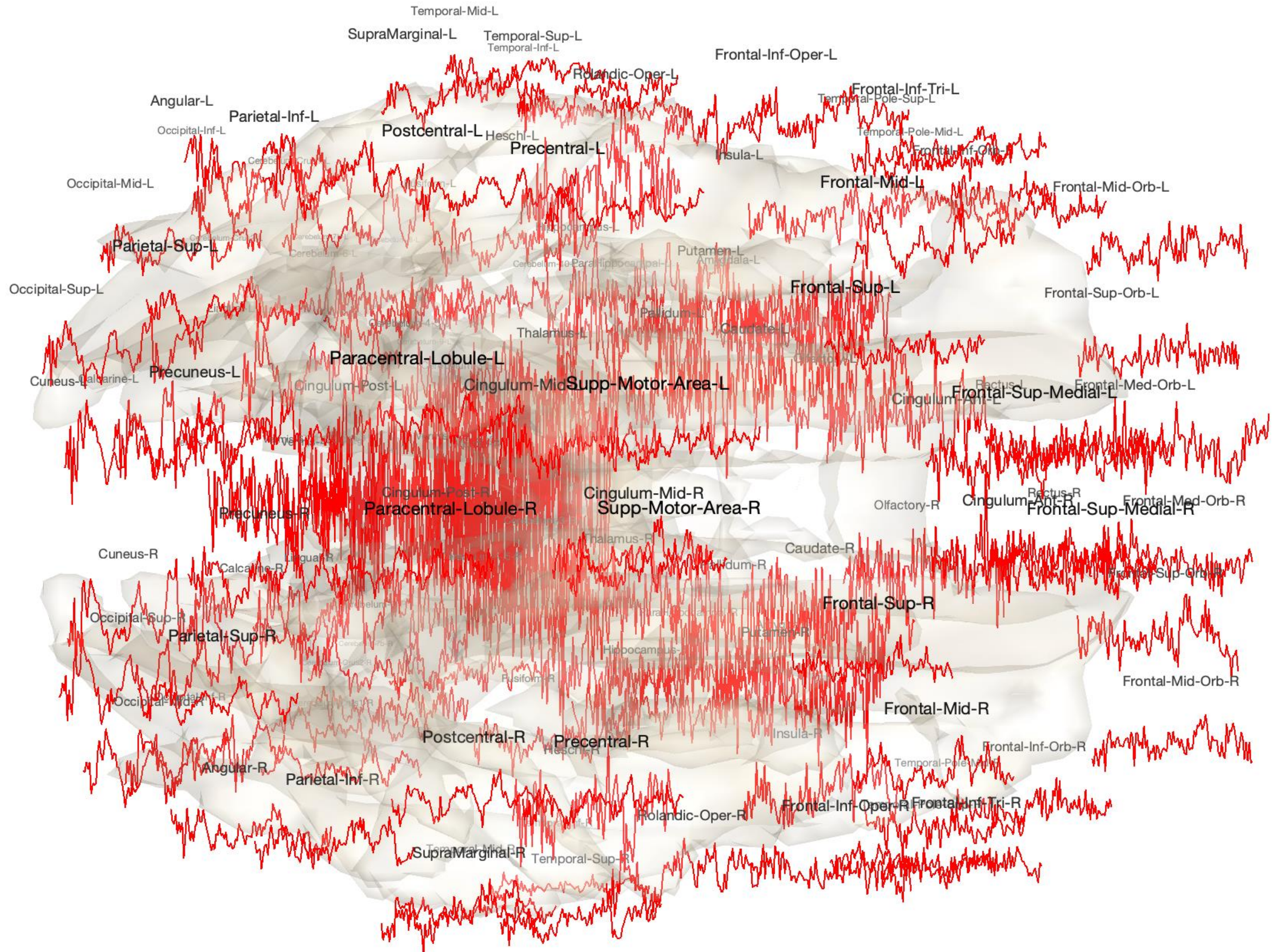
3T GE Discovery MR750
Center for Imaging Research
Medical College of
Wisconsin, Milwaukee, WI

rs-fMRI (every 30 second)

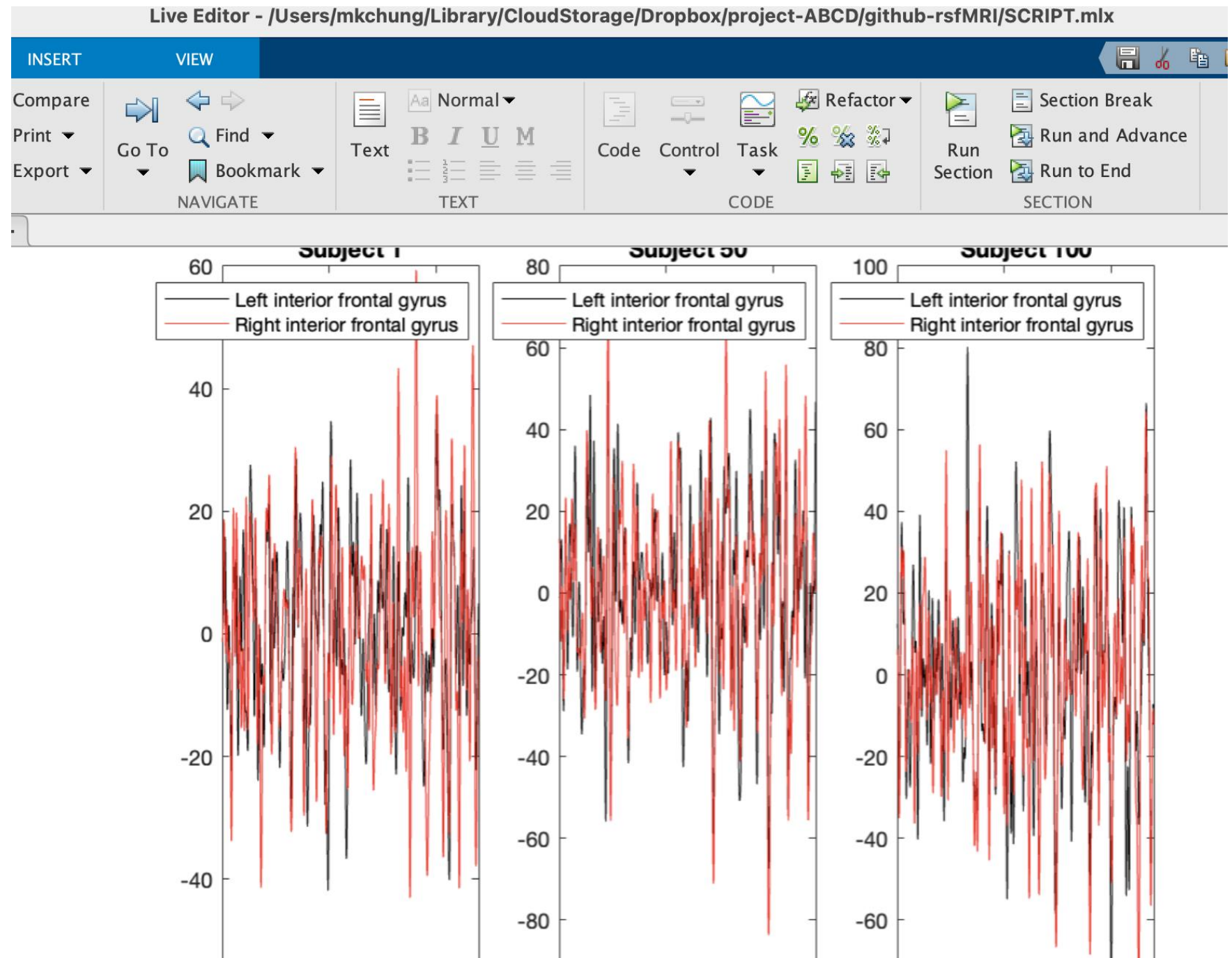


Time series averaged into 116 brain

Y



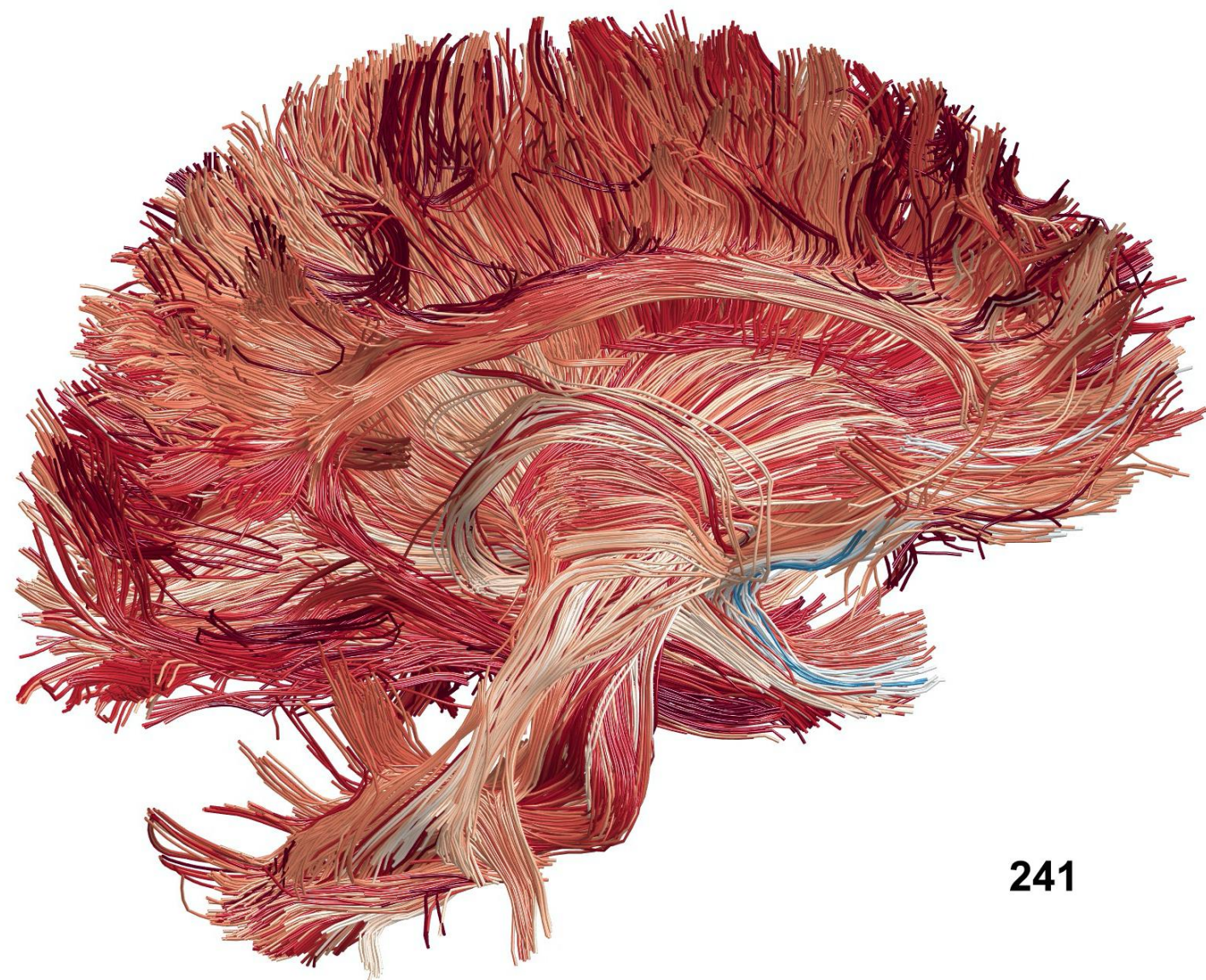
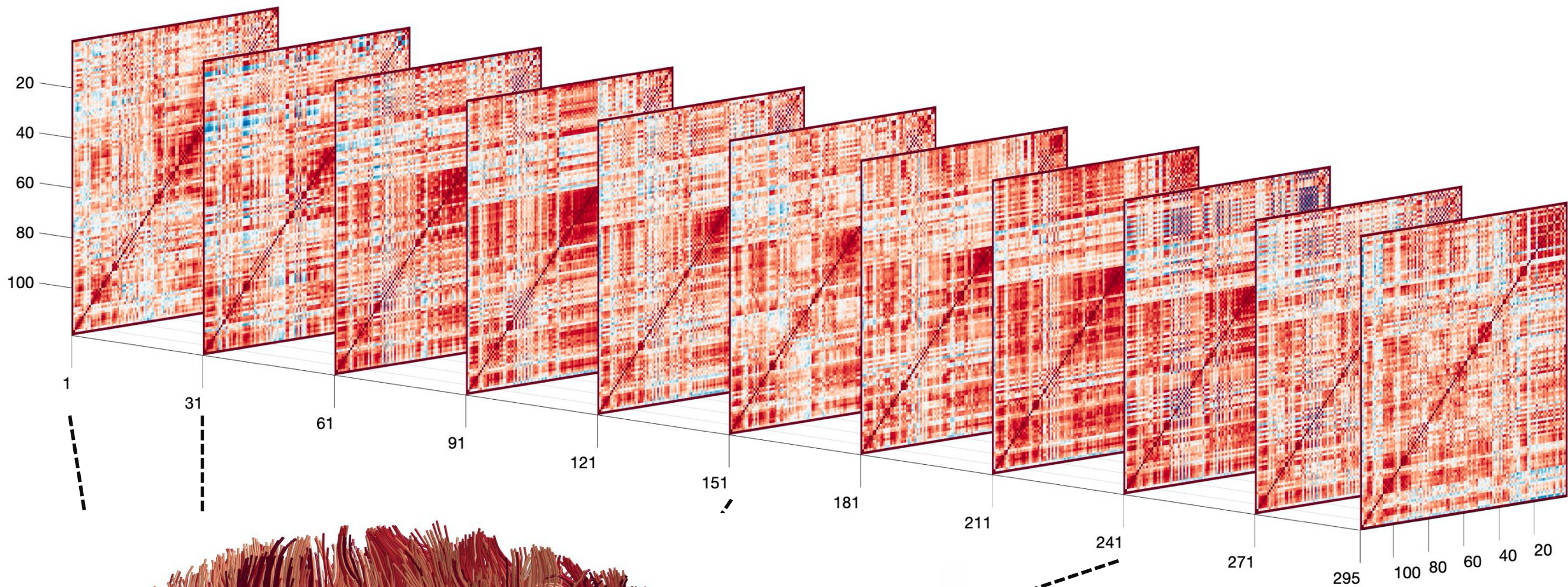
Sample rs-fMRI time series data / rsfMRI



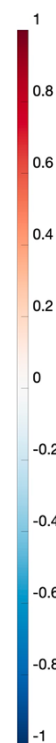
*Important
biological
questions are*

added

Huang et al. 2020 Neuroscience Methods 331:108480



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Dynamically
changing functional
brain network
superimposed on top
of white matter fibers

Distance between clusters

$$C_1 \cup C_2 \cup \cdots \cup C_k = \{\mathcal{X}_1, \cdots, \mathcal{X}_n\}, \quad C_i \cap C_j = \emptyset$$

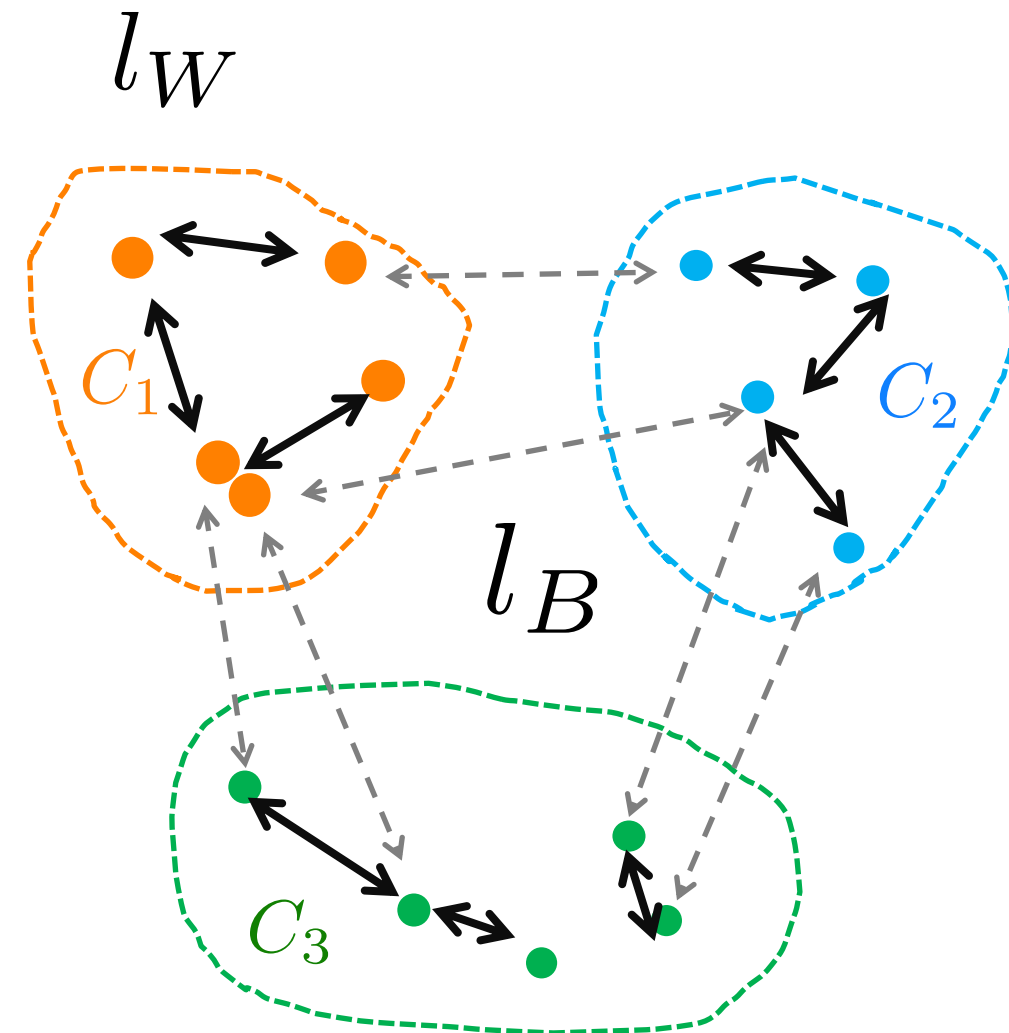
Between-group distance

$$l_B \propto \sum_{i \in C_1, j \in C_2} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j) \quad \leftarrow \text{---} \text{ 0D and 1D combined distances}$$

Within-group distance

$$l_W \propto \sum_k \sum_{i, j \in C_k} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

$$l_B + l_W = \sum_{i, j} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$



Clustering

Minimize the within cluster distance over all possible clustering configurations

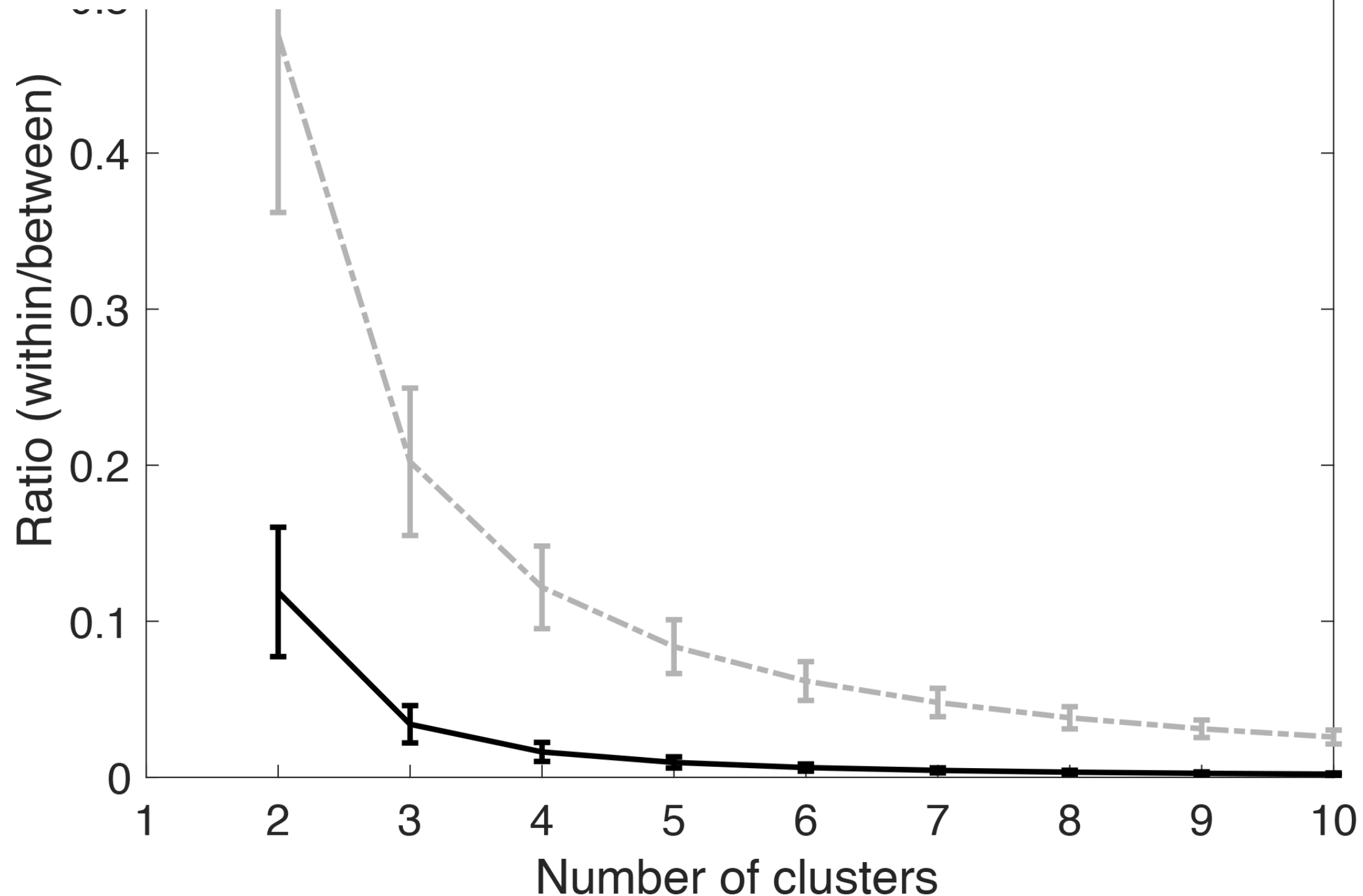
$$l_W \propto \sum_k \sum_{i,j \in C_k} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

Equivalently, maximize the between cluster distance

$$l_B = \sum_{i \in C_1, j \in C_2} \mathcal{L}(\mathcal{X}_i, \mathcal{X}_j)$$

Elbow method: Optimal number of clusters

$$\frac{1}{\phi} = \frac{l_W}{l_B}$$



The within cluster variance *6 times* smaller

Self exercise problem

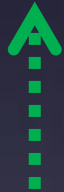
Investigate how clustering accuracy is computed.

Clustering accuracy measures how well the estimated cluster assignments match the true labels. Because clustering is obtained through data-driven estimation rather than deterministic labeling, the reported accuracy reflects **statistical estimation performance** rather than an exact or deterministic quantity.

Mathematical equivalence of topological clustering and topological inference

There exists a monotonically decreasing function f satisfying

$$p\text{-value} = f(\text{clustering accuracy})$$


Inference


Clustering

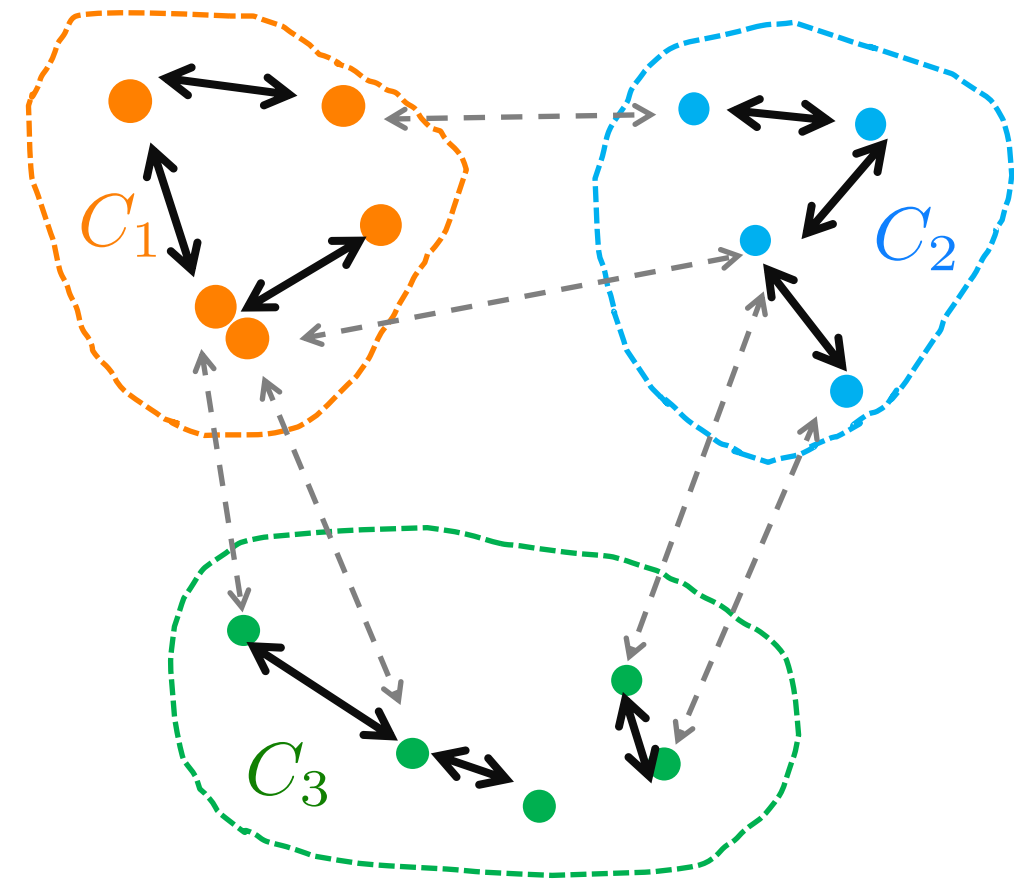
Topological clustering = topological inference

Within-group distance $d_W \propto \sum_k \sum_{i,j \in C_k} d_0(\mathcal{X}_i, \mathcal{X}_j) + d_1(\mathcal{X}_i, \mathcal{X}_j)$

Clustering accuracy

$$\frac{dA}{dp} = \frac{dA}{d(d_W)} \frac{d(dl_W)}{dp} \leq 0$$

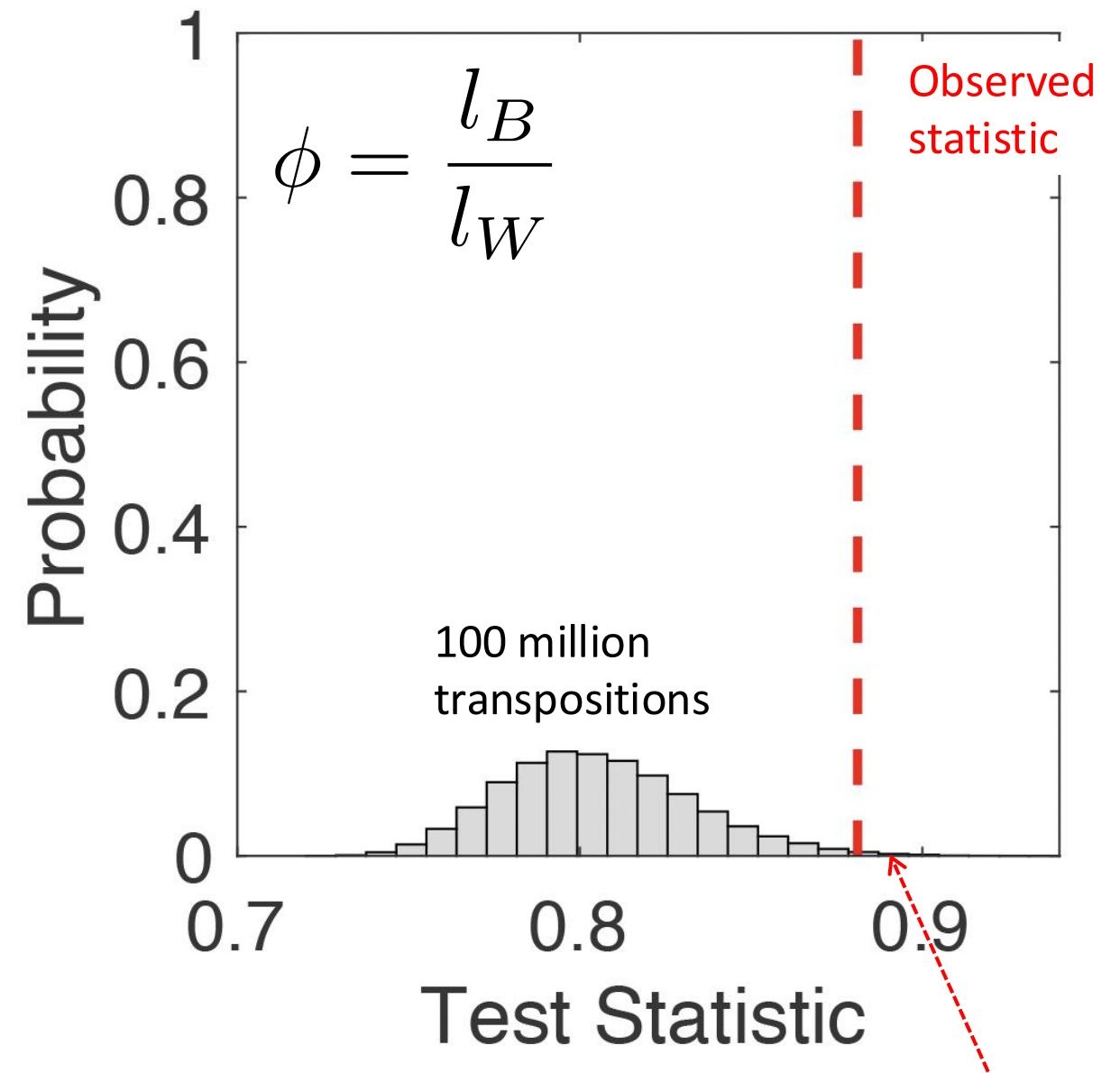
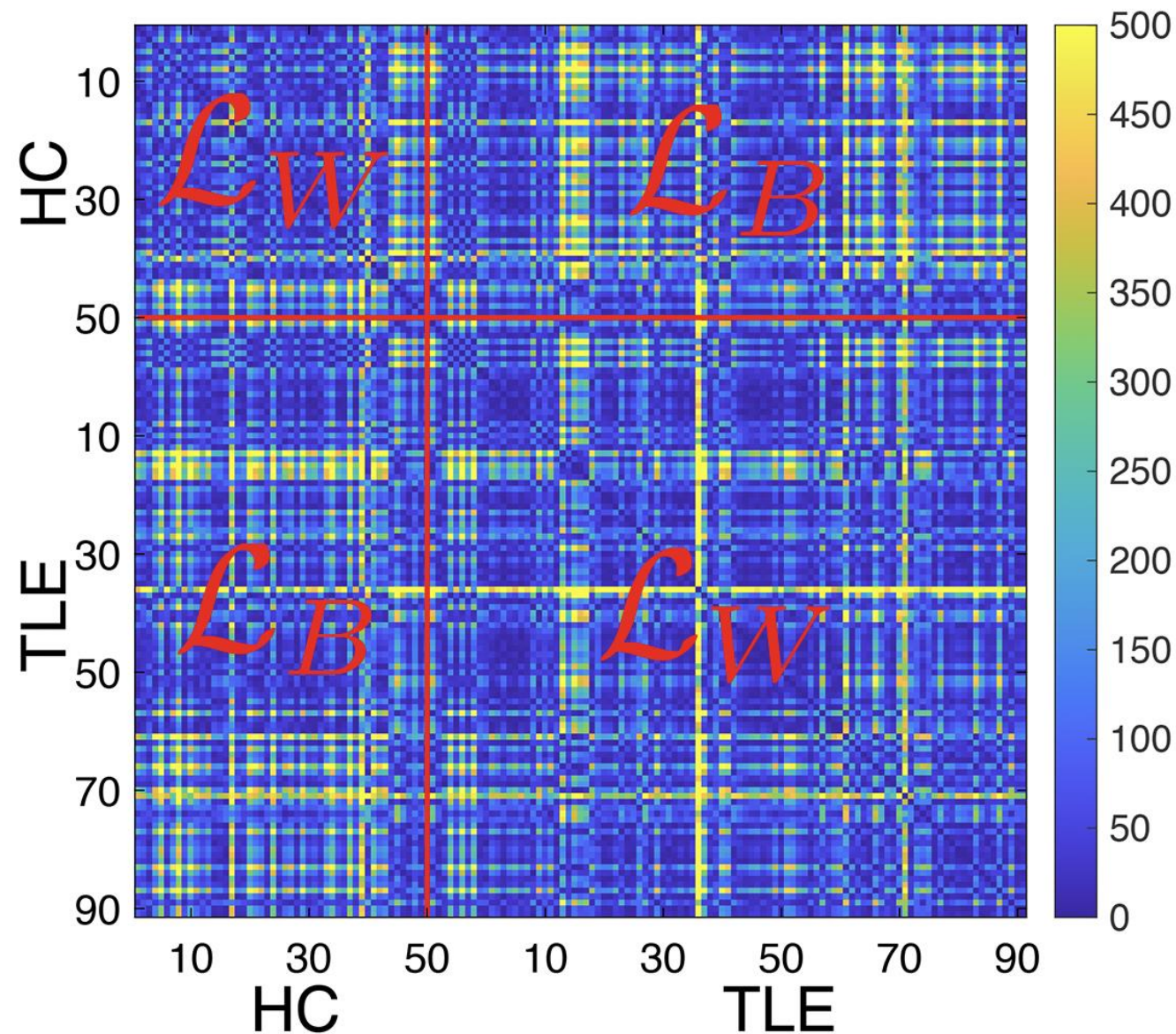
p -value



There exists a **monotone decreasing** function f satisfying $p = f(A)$

Sketch of Proof in Chung et al. 2023 Neurolmage

Statistical inference on the equivalence of clusters



P-value 0.0086

$$l_W \rightarrow l_W + \Delta(\text{transposition})$$

$$l_B \rightarrow l_B + \Delta(\text{transposition})$$

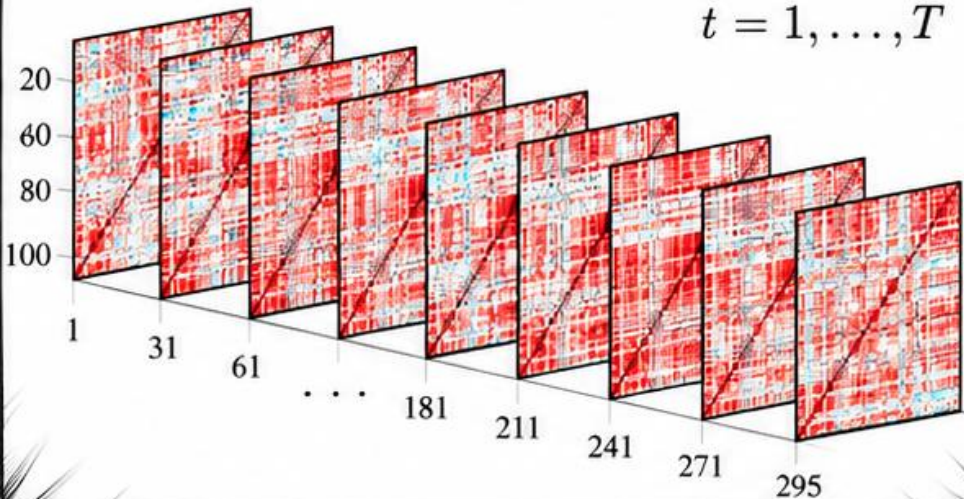
Songdechakraiwut and Chung
2023

k-means clustering to state space estimation

1. Time-varying connectivity matrices

For each subject, compute a sequence of connectivity matrices over time.

$$\mathbf{A}^{(t)} \in \mathbb{R}^{n \times n}$$
$$t = 1, \dots, T$$



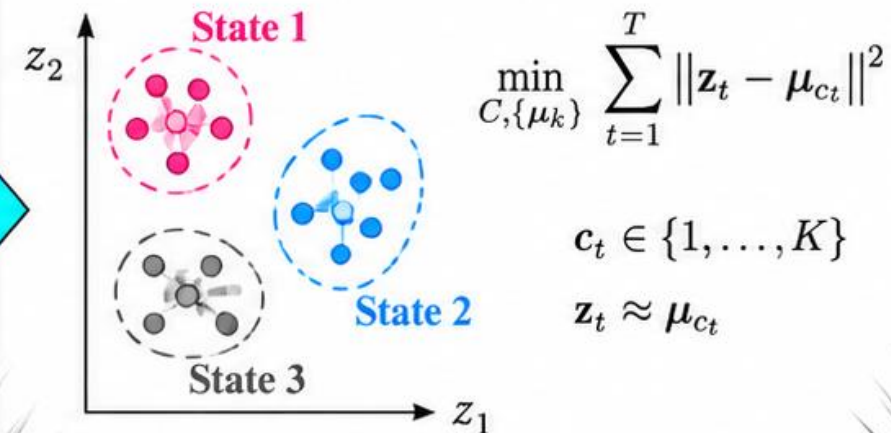
2. Vectorize and normalize

Convert each matrix to a feature vector (using the upper triangle) and normalize.

$$\mathbf{A}^{(t)}$$
$$\mathbf{x}_t = \text{vec}(\mathbf{A}_{\text{upper}}^{(t)})$$
$$\in \mathbb{R}^d$$
$$d = n(n-1)/2$$
$$\mathbf{z}_t = \frac{\mathbf{x}_t - \boldsymbol{\mu}}{\|\mathbf{x}_t - \boldsymbol{\mu}\|}$$

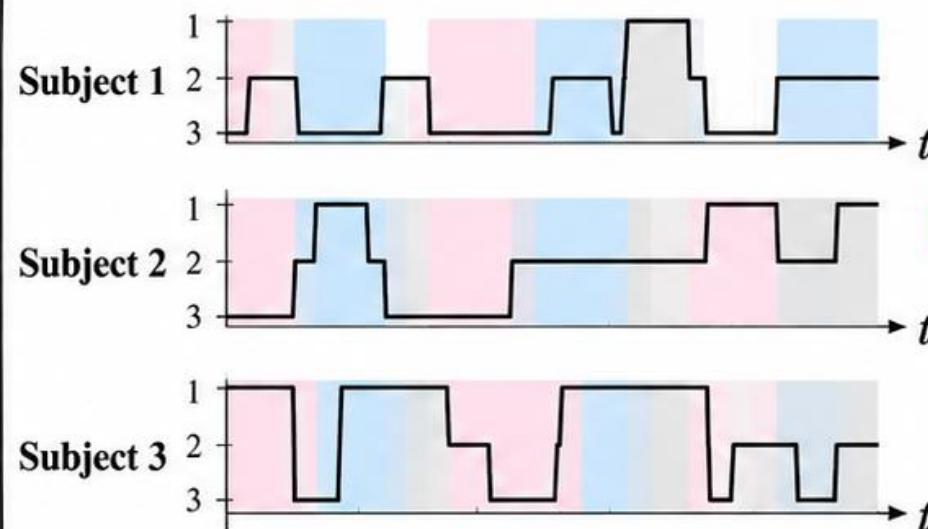
3. *k*-means clustering in state space

Cluster the normalized vectors $\{\mathbf{z}_t\}_{t=1}^T$ into K states using *k*-means.



4. State sequence for each subject

Assign each time point to the nearest cluster centroid.

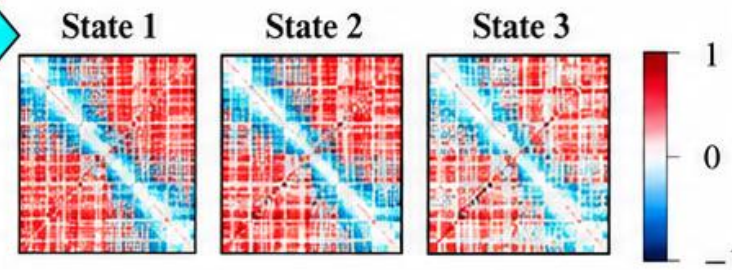


Each subject has a sequence of brain states over time.

5. State-specific connectivity

For each state k , compute the mean matrix of all time points assigned to that state.

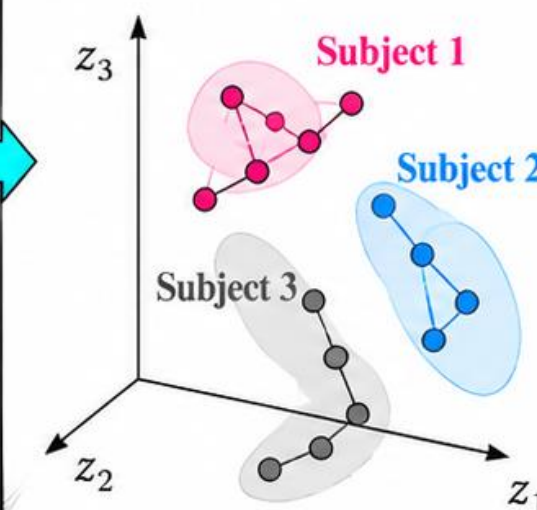
$$\bar{\mathbf{A}}^{(k)} = \frac{1}{|S_k|} \sum_{t \in S_k} \mathbf{A}^{(t)}$$



Each state represents a characteristic connectivity pattern.

6. State space and further analysis

Represent each subject by a trajectory in a low-dimensional state space and build a statistical model.

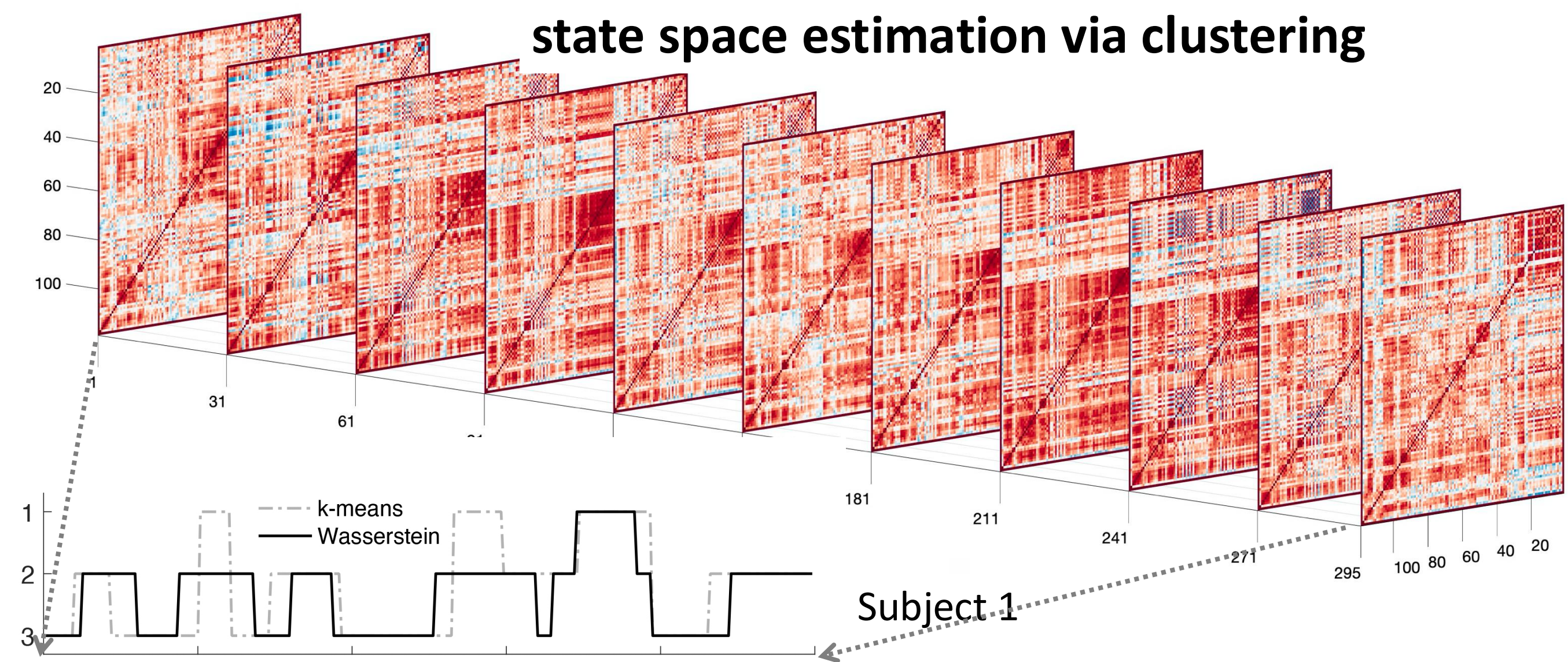


Applications:

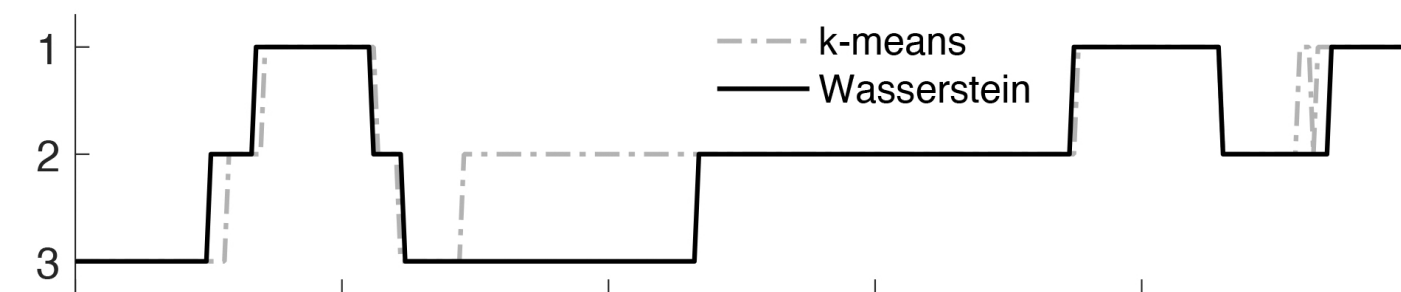
- compare state dynamics across subjects
- group-level modeling
- relate states to behavior or covariates

Problem: Is the state reproducible biomarker?

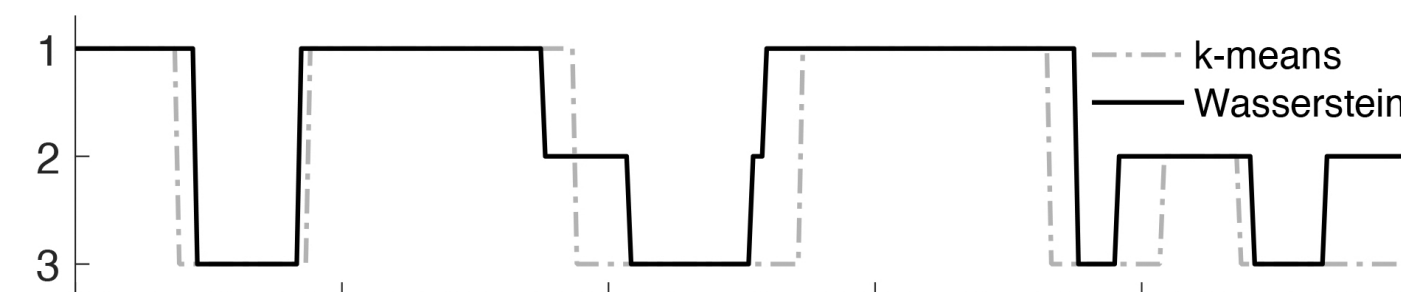
state space estimation via clustering



Subject 1



Subject 2

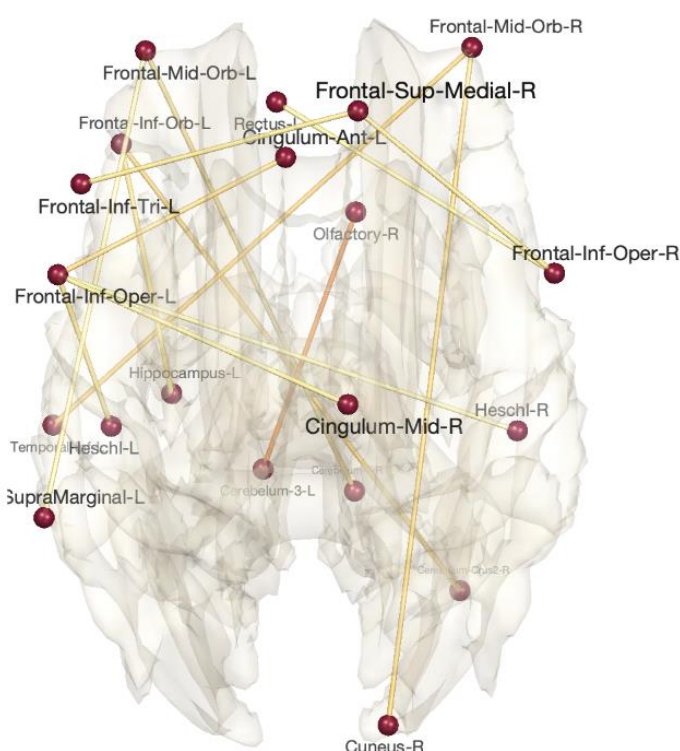
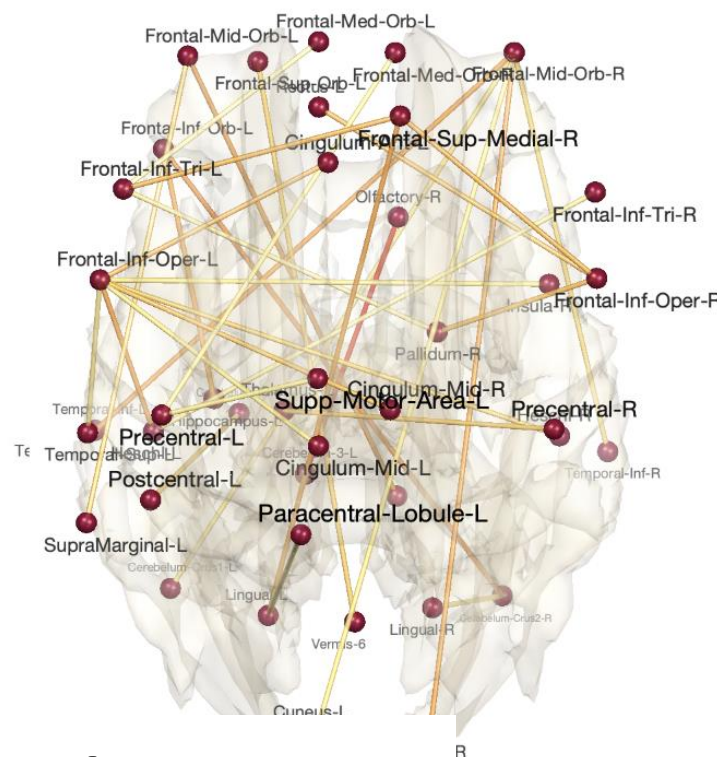
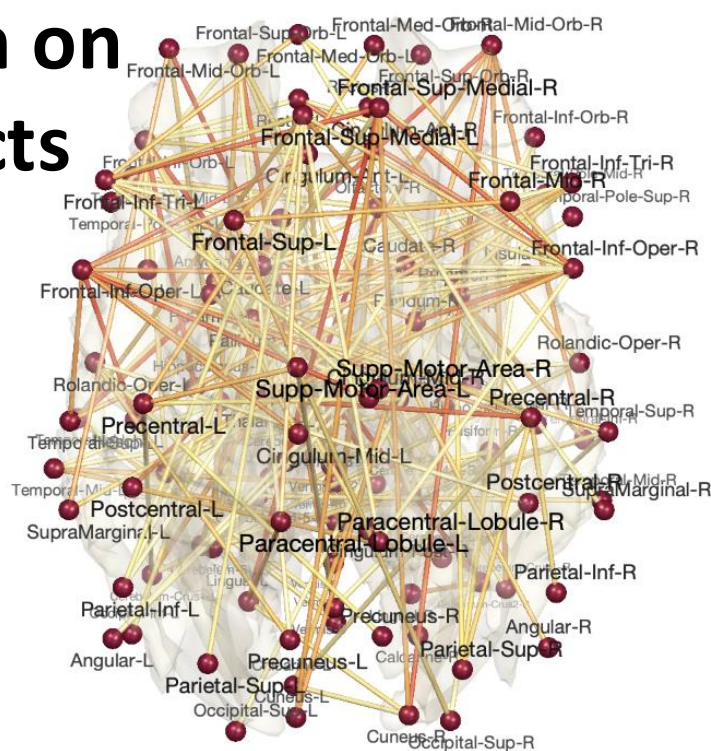


Subject 3

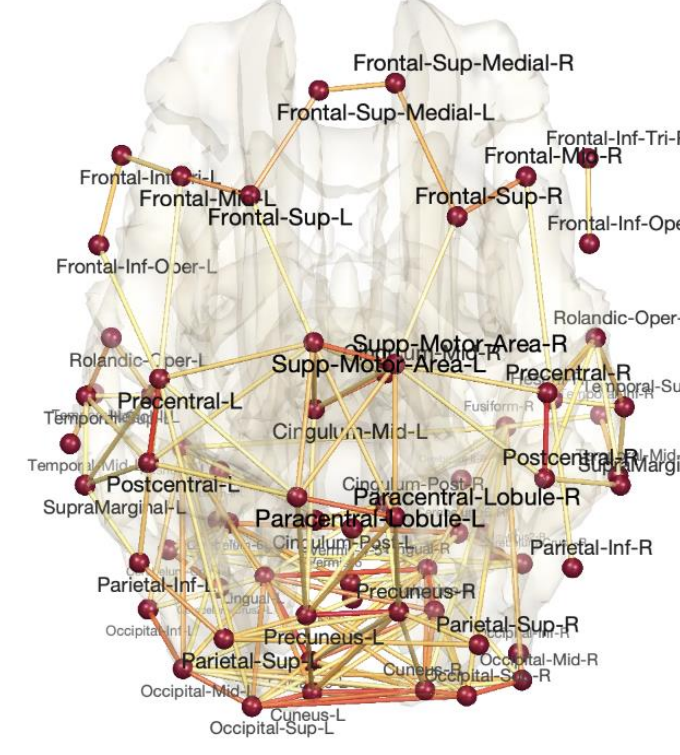
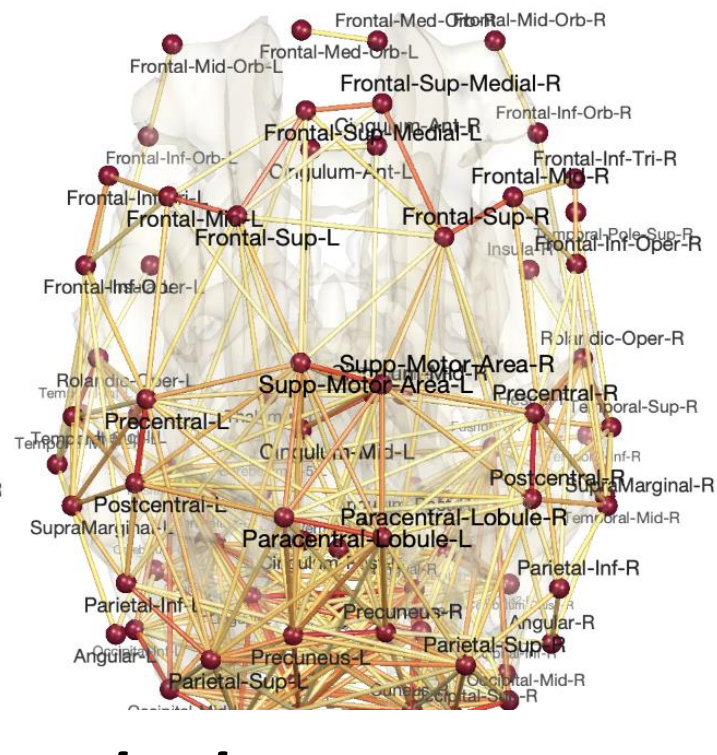
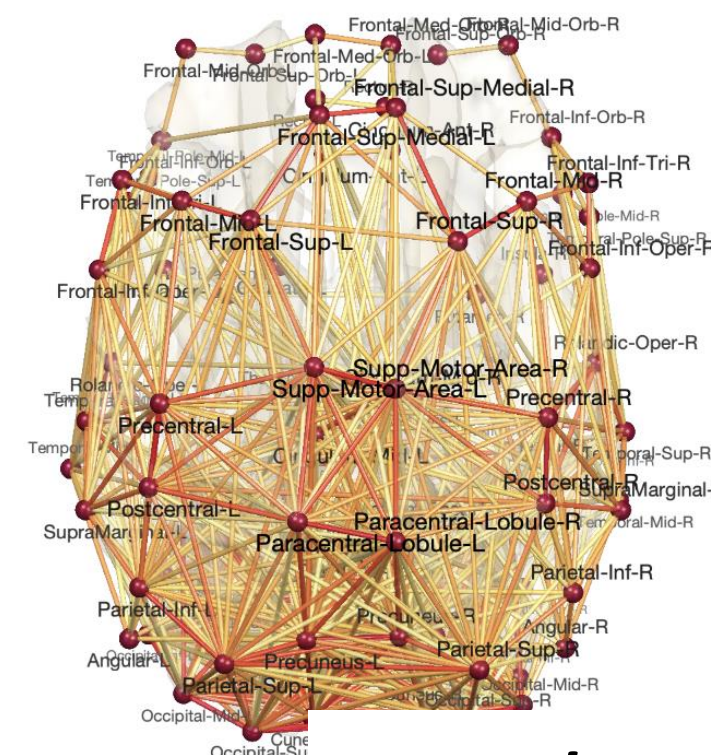
2nd level
analysis:

Statistical model

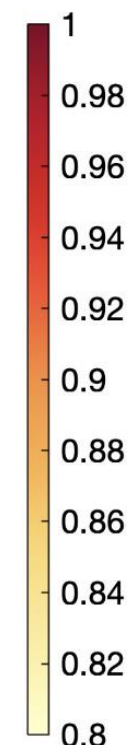
State space estimation on 479 subjects



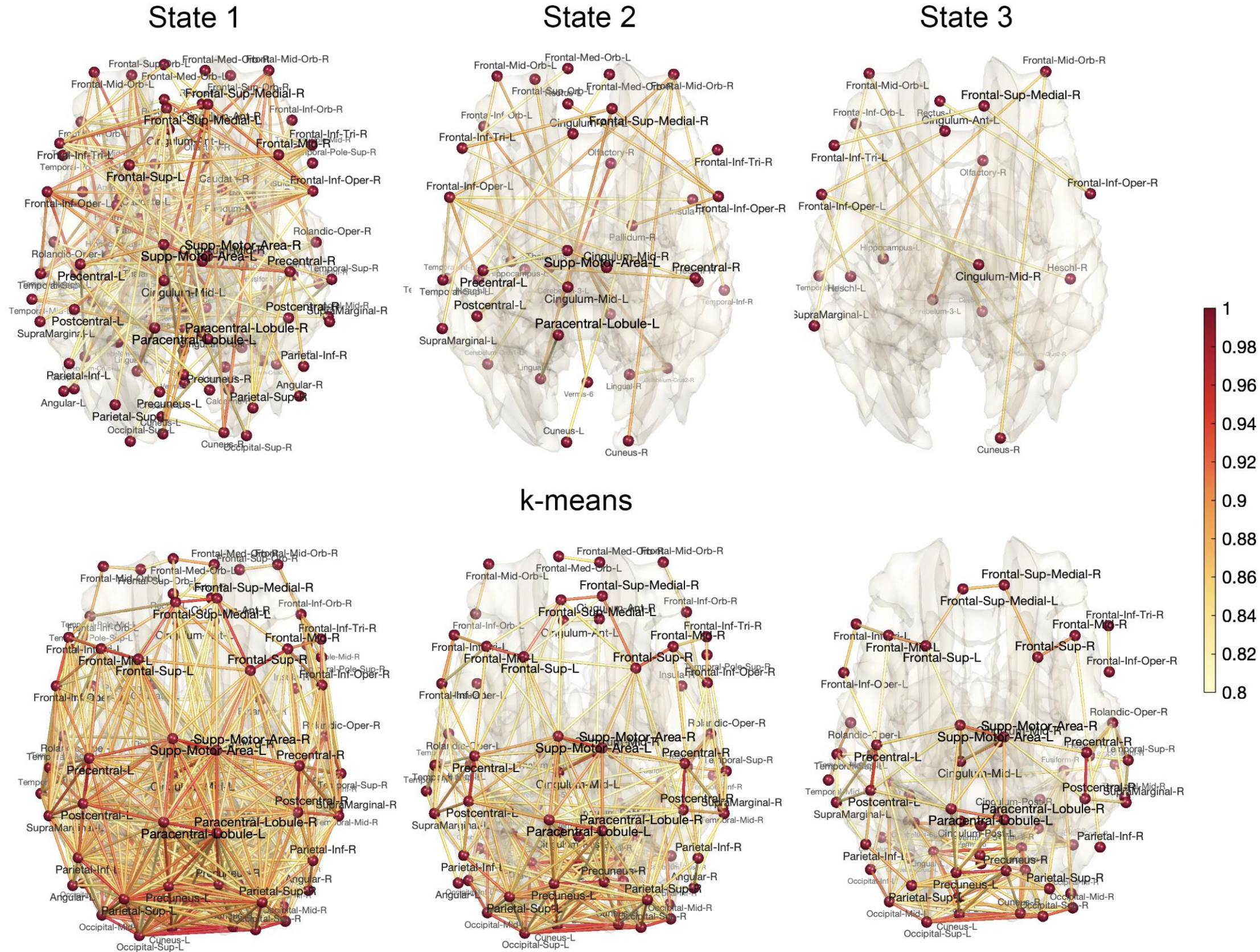
k-means Sample mean in each state



Topological clustering Topological mean in each state

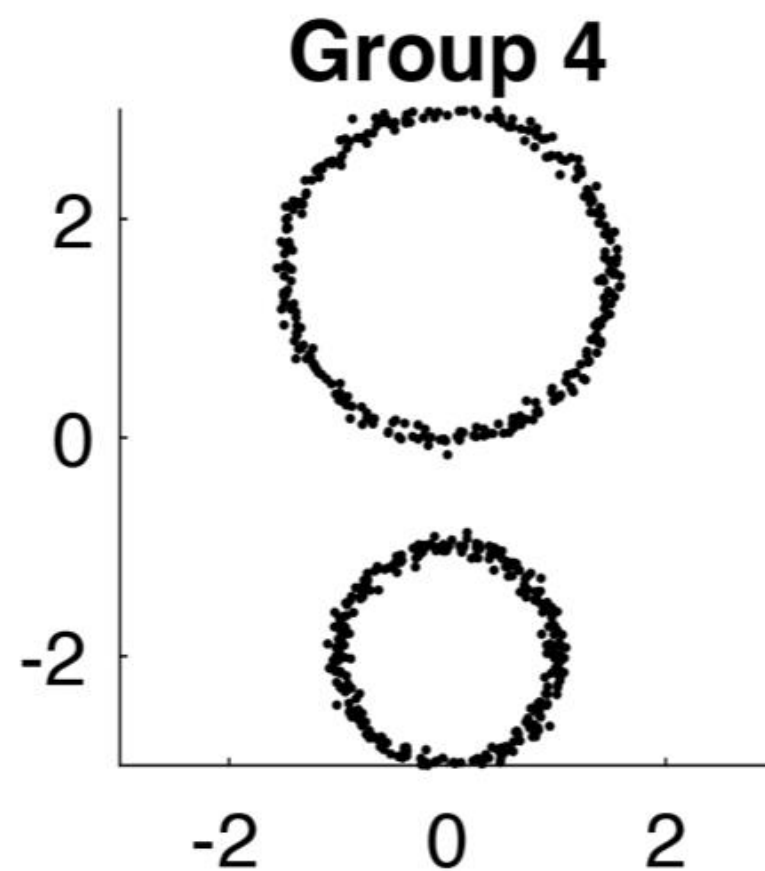
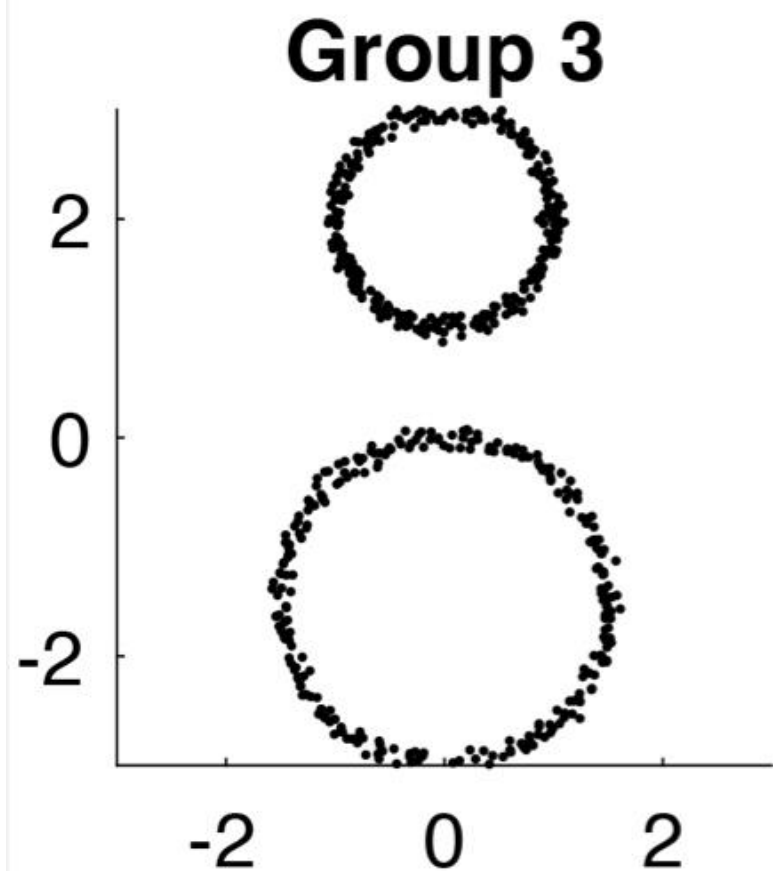
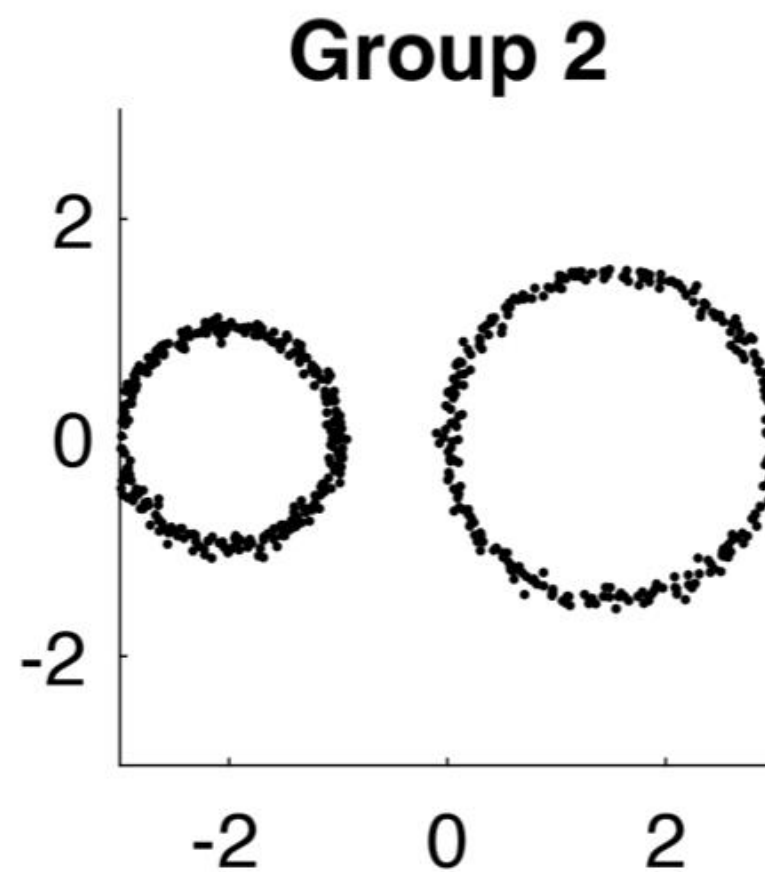
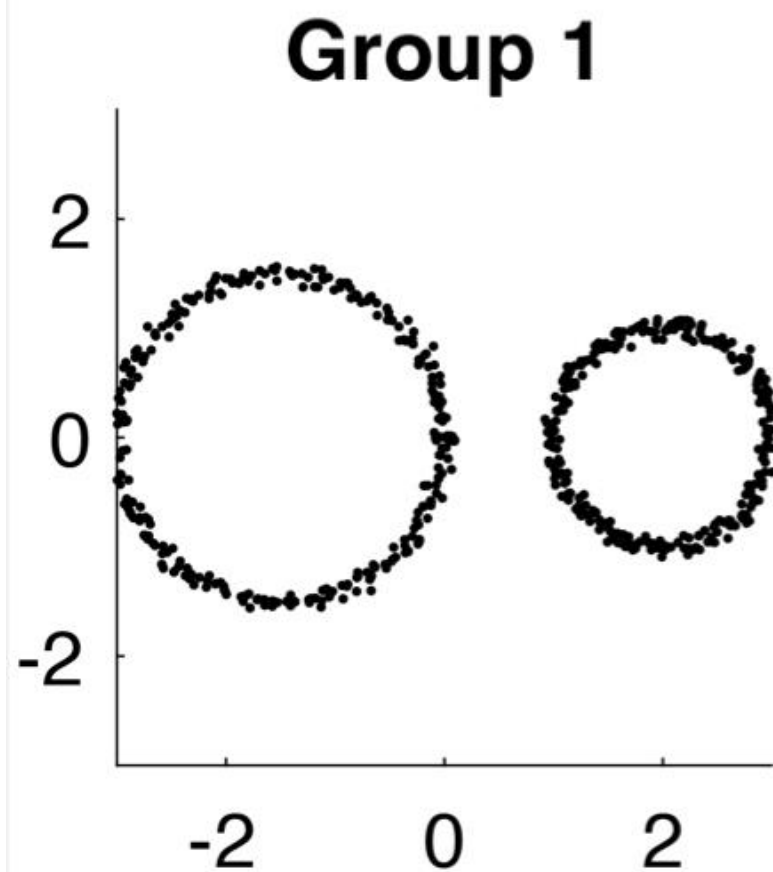


But which method is more accurate or correct?



Often need validation on ground truth toy examples

Geometric clustering methods fail topological task



*All false
positives*

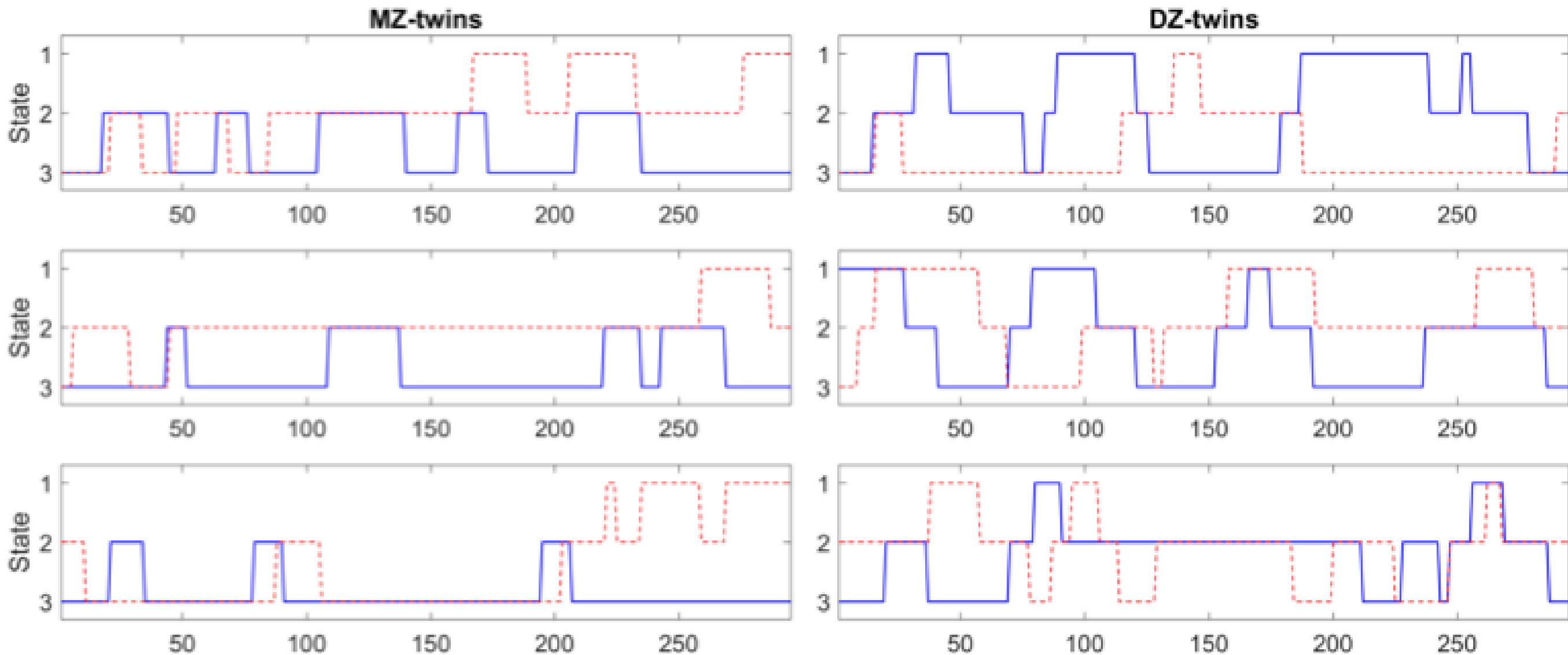
K-means
clustering
 0.98 ± 0.01

Hierarchical
clustering
 1.00 ± 0.00

Topological
clustering
 0.63 ± 0.04



Is the state-change heritable?



UW-Madison twin study (200 twin pairs)

Topological clustering can answer this question

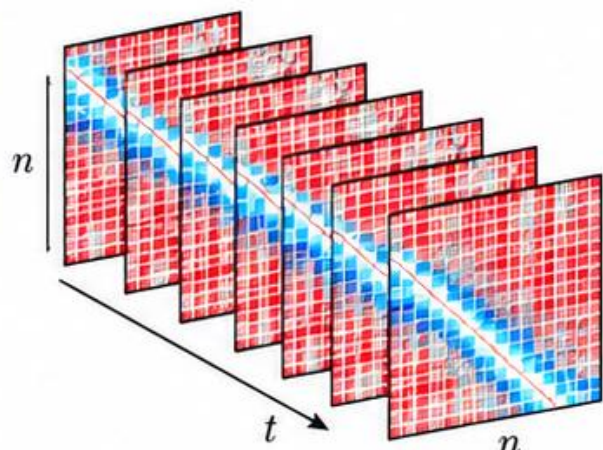
Chung et al. 2024 [arXiv:2201.00087](https://arxiv.org/abs/2201.00087) (PLOS Computational

What is wrong with the state space estimation?

1 Time-varying connectivity

For each subject s , compute a sequence of connectivity matrices

$$A_s(t) \in \mathbb{R}^{n \times n}, \quad t = 1, \dots, T_s$$



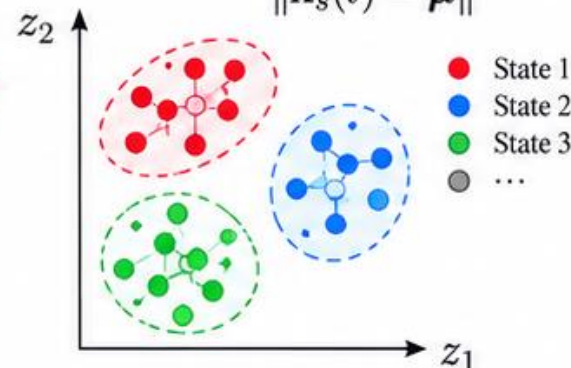
(e.g., correlation, coherence, etc.)

2 Clustering to obtain states

Vectorize and normalize each matrix

$$\mathbf{x}_s(t) = \text{vec}(A_s(t)_{\text{upper}}) \in \mathbb{R}^d$$

$$\mathbf{z}_s(t) = \frac{\mathbf{x}_s(t) - \boldsymbol{\mu}}{\|\mathbf{x}_s(t) - \boldsymbol{\mu}\|}$$

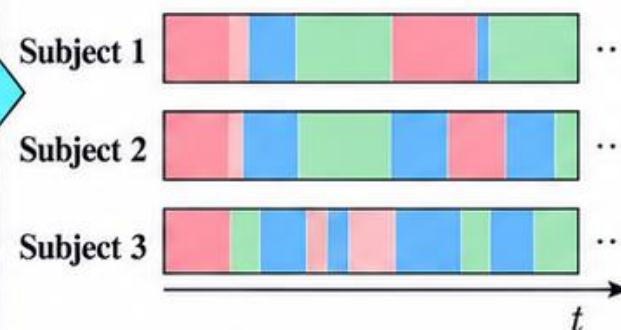


Apply k -means to $\{\mathbf{z}_s(t)\}$ from all subjects to obtain K cluster centroids $\{\boldsymbol{\mu}_k\}$.

3 State sequence for each subject

Assign each time point to the nearest centroid:

$$c_s(t) = \arg \min_k \|\mathbf{z}_s(t) - \boldsymbol{\mu}_k\|$$

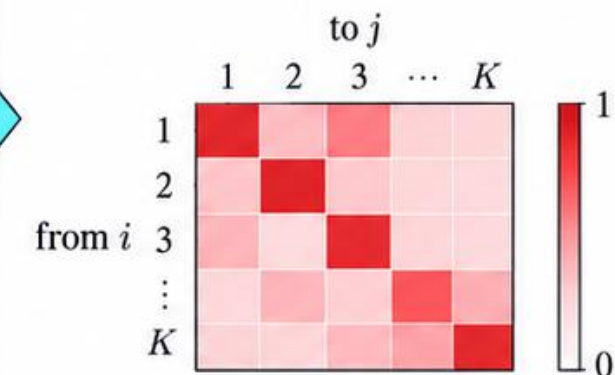


Each subject yields a discrete state sequence $\{c_s(t)\}_{t=1}^{T_s}$, $c_s(t) \in \{1, \dots, K\}$.

4 Subject-level dynamics

Model the state sequence as a Markov process (or HMM):

$$P_{ij}^{(s)} = \Pr(c_s(t+1) = j \mid c_s(t) = i)$$



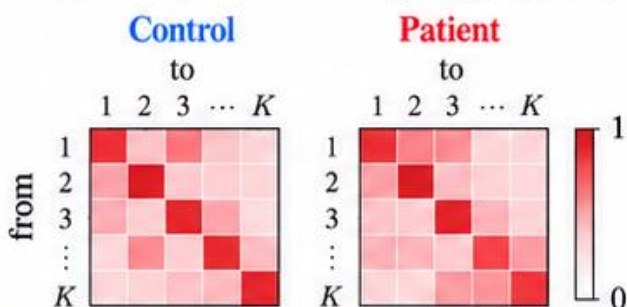
Estimate $P^{(s)}$ from counts or using HMM to allow for noisy observations.

5 Group-level model across subjects

Model subject variability and covariate effects, e.g., multinomial regression:

$$P_{ij}^{(s)} = \frac{\exp(\beta_{ij}^T \mathbf{y}_s)}{\sum_{l=1}^K \exp(\beta_{il}^T \mathbf{y}_s)}$$

$\mathbf{y}_s$: subject covariates (group, age, etc.)

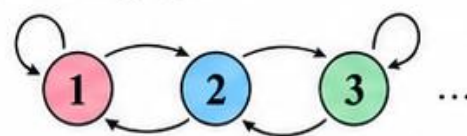


Test group differences in transitions and/or state occupancies.

6 Additional state dynamics

Other useful quantities:

- **Occupancy rate:** $\pi_s(k) = \frac{1}{T_s} \sum_t \mathbf{1}(c_s(t) = k)$
- **Dwell time:** distribution of consecutive time points in the same state
- **Transition graph:**



- **Continuous-time model (optional):**

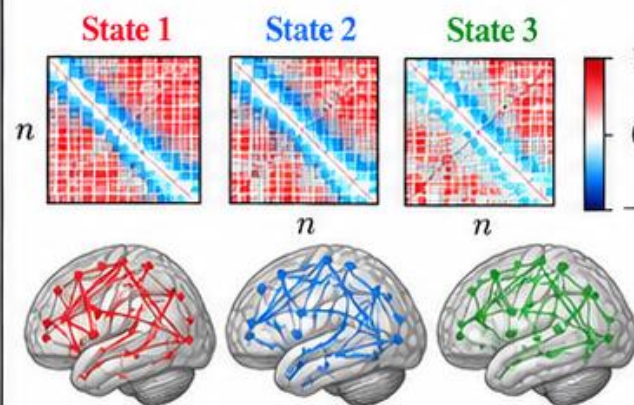
$$\frac{d\mathbf{p}_s(t)}{dt} = \mathbf{p}_s(t) \mathbf{Q}^{(s)}$$

(generator matrix $\mathbf{Q}^{(s)}$)

7 State-specific brain patterns

For each state k , average the matrices assigned to that state:

$$\bar{A}^{(k)} = \frac{1}{\sum_s |T_{s,k}|} \sum_s \sum_{t \in T_{s,k}} A_s(t)$$

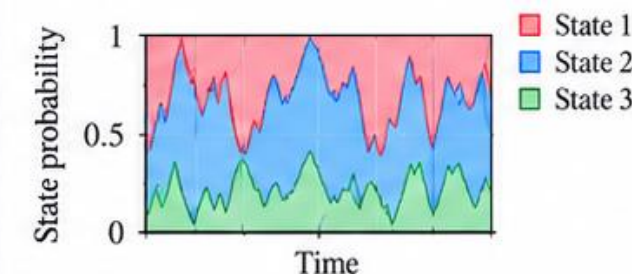


Each state corresponds to a characteristic brain connectivity pattern.

8 Analysis and interpretation

Use the learned state dynamics to:

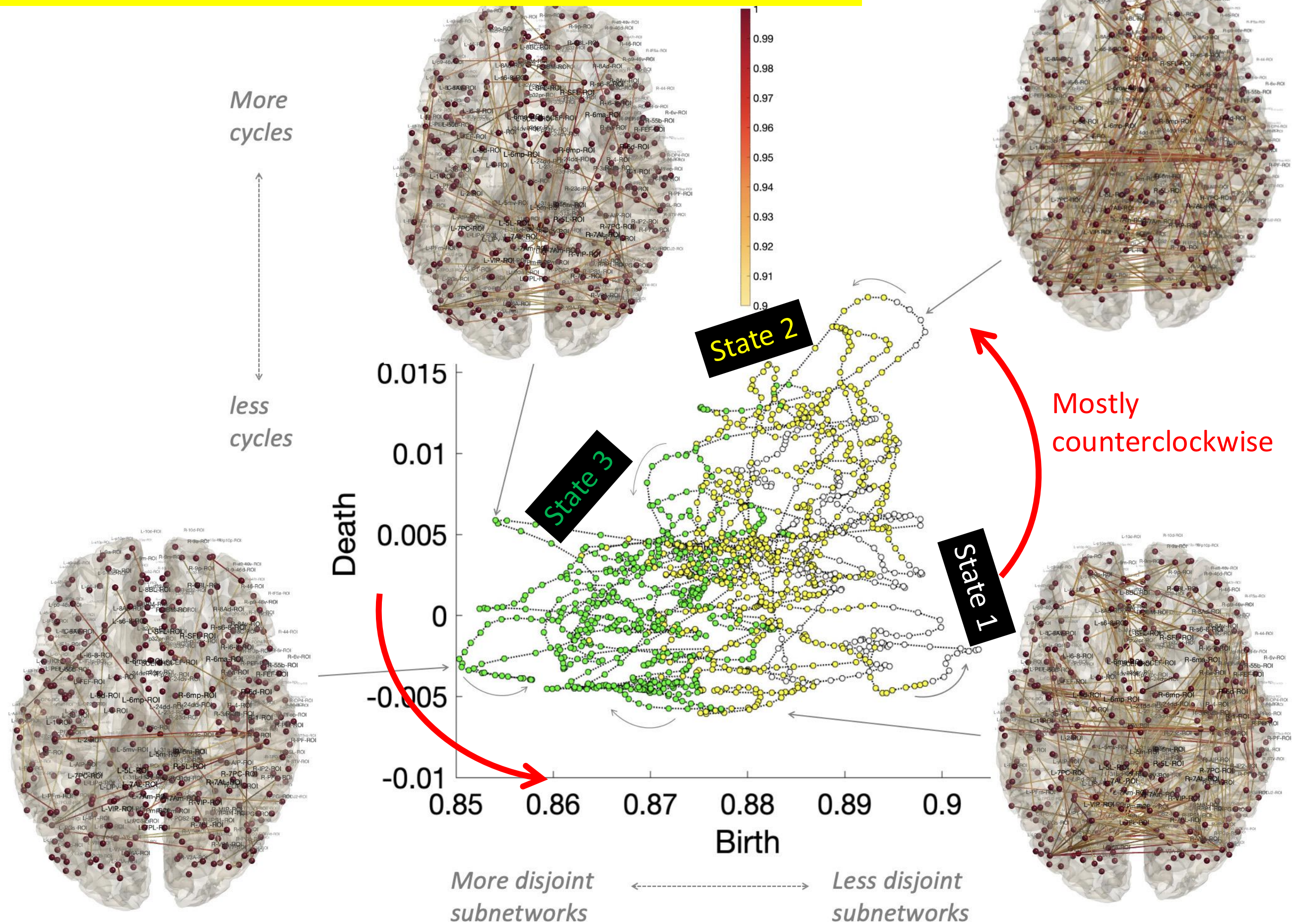
- Compare groups (e.g., control vs. patient)
- Relate transitions/occupancies to behavior or clinical variables
- Predict subject-level trajectories
- Identify biomarkers (e.g., state features, dynamics parameters)



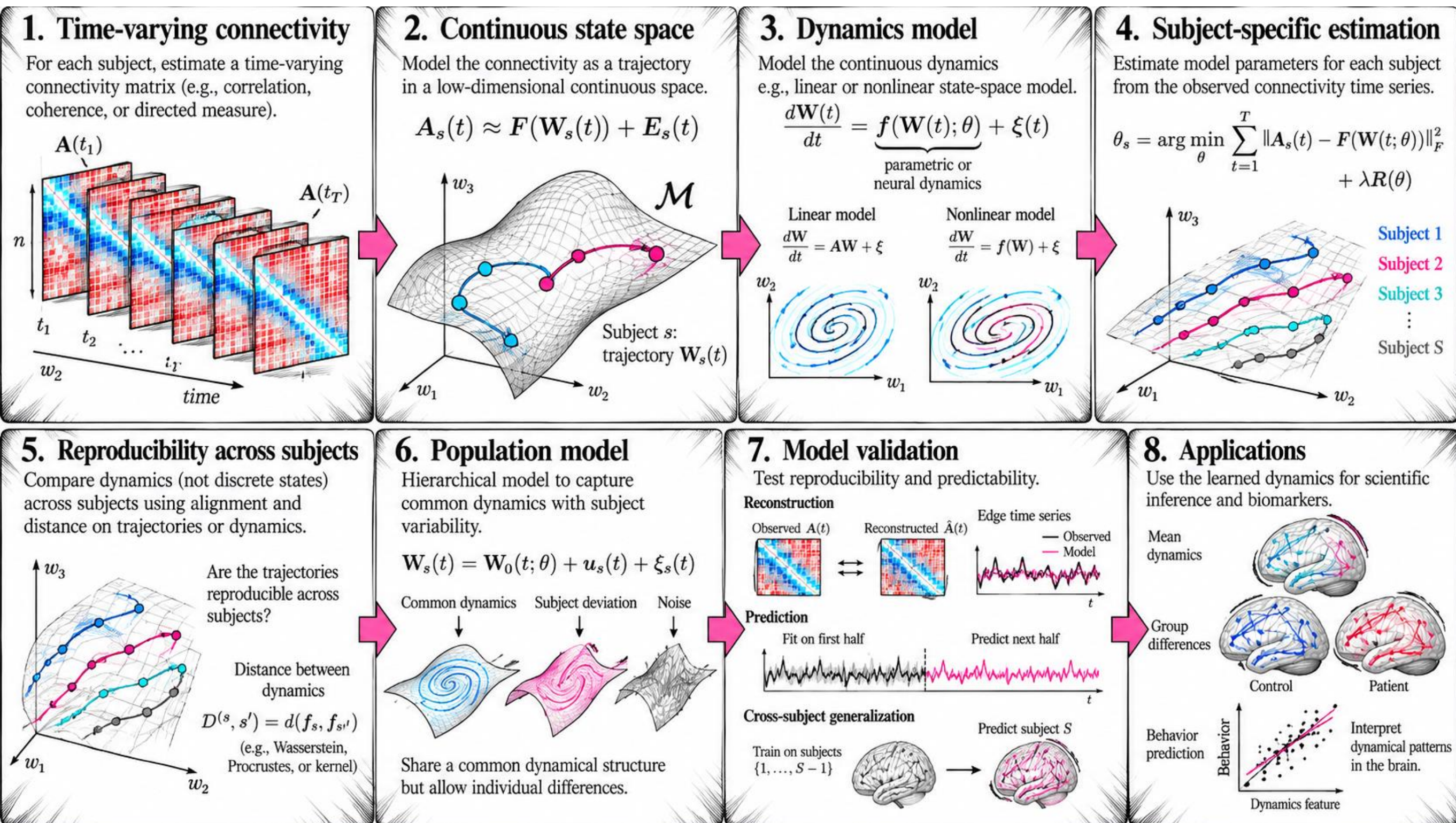
Result: a principled, generative model of brain state dynamics across subjects.

Project: Determine if the state change dynamics itself is reproducible.

Interpreting clustering results is often needed



Better approach: Dynamic state-space model



Project: Determine if the state change dynamics itself is reproducible.

Learning outcome

Can formulate clustering problem

Can formulate topological clustering

Can understand the performance difference
between topological vs. geometric clustering